

Q1 Finite Elements And Stiffness Assembly

Compact cheatsheet with course notation

1 Q1 Reference Element

Course notation: $K = [0, 1]^2$, $P = \text{span}(1, x, y, xy)$, and $\Sigma = \{p \rightarrow p(0, 0), p \rightarrow p(1, 0), p \rightarrow p(0, 1), p \rightarrow p(1, 1)\}$.

Node ordering: $a_1 = (0, 0)$, $a_2 = (1, 0)$, $a_3 = (0, 1)$, $a_4 = (1, 1)$.

Local basis functions: $\varphi_1 = (1-x)(1-y)$, $\varphi_2 = x(1-y)$, $\varphi_3 = (1-x)y$, $\varphi_4 = xy$. They satisfy $\varphi_{i(a_j)} = \delta_{ij}$, hence every $p \in Q_1$ satisfies $p(x, y) = \sum_{i=1}^4 p(a_i) \varphi_{i(x,y)}$.

Interpretation: Q_1 is the tensor product of two one-dimensional P_1 elements. It is affine in each variable separately, but not globally affine because of the xy term.

2 Global Q1 Space

On a rectangular mesh $(K_l)_{1 \leq l \leq N_t}$ of Ω , define $V_h = \{v_h \in C^0(\bar{\Omega}) : v_h|_{K_l} \in Q_1, \forall l\}$. If the mesh has N_v vertices, then $\dim V_h = N_v$.

The global basis functions Φ_i satisfy $\Phi_{i(a_j)} = \delta_{ij}$ for all global vertices a_j . For homogeneous Dirichlet boundary conditions, use $V_{h,0} = V_h \cap H_0^1(\Omega) = \text{span} \{\Phi_i : a_i \text{ is an interior vertex}\}$.

3 Model Problem And Matrix Form

For $-\Delta u = f$ in Ω , $u = 0$ on $\partial\Omega$, the variational formulation is

$$\text{Find } u \in H_0^1(\Omega) \text{ such that } \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega).$$

The discrete formulation is the same with $H_0^1(\Omega)$ replaced by $V_{h,0}$. Writing $u_h = \sum_j U_j \Phi_j$ gives $AU = F$, with $A_{ij} = \int_{\Omega} \nabla \Phi_i \cdot \nabla \Phi_j \, dx$ and $F_i = \int_{\Omega} f \Phi_i \, dx$. The matrix A is the global stiffness matrix.

4 Reference Local Stiffness Matrix

On $K = [0, 1]^2$, define $K_{ij}^{\text{ref}} = \int_K \nabla \varphi_j \cdot \nabla \varphi_i \, dx \, dy$. With the node ordering $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$:

$$K^{\text{ref}} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{6} & -\frac{1}{6} & -\frac{1}{3} \\ -\frac{1}{6} & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{6} & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{3} & -\frac{1}{6} & -\frac{1}{6} & \frac{2}{3} \end{pmatrix}.$$

For rectangular cells, keep the derivative split. Define $K_x^{\text{ref}_{ij}} = \int_K \partial_x \varphi_j \partial_x \varphi_i \, dx \, dy$ and $K_y^{\text{ref}_{ij}} = \int_K \partial_y \varphi_j \partial_y \varphi_i \, dx \, dy$. Then $K^{\text{ref}} = K_x^{\text{ref}} + K_y^{\text{ref}}$, with

$$K_x^{\text{ref}} = \begin{pmatrix} \frac{1}{3} & -\frac{1}{3} & \frac{1}{6} & -\frac{1}{6} \\ -\frac{1}{3} & \frac{1}{3} & -\frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & -\frac{1}{6} & \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{6} & \frac{1}{6} & -\frac{1}{3} & \frac{1}{3} \end{pmatrix}, \quad K_y^{\text{ref}} = \begin{pmatrix} \frac{1}{3} & \frac{1}{6} & -\frac{1}{3} & -\frac{1}{6} \\ \frac{1}{6} & \frac{1}{3} & -\frac{1}{6} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{6} & \frac{1}{3} & \frac{1}{6} \\ -\frac{1}{6} & -\frac{1}{3} & \frac{1}{6} & \frac{1}{3} \end{pmatrix}.$$

5 Physical Rectangle

Let $K_l = [x_i, x_{i+1}] \times [y_j, y_{j+1}]$, with $h_x = x_{i+1} - x_i$ and $h_y = y_{j+1} - y_j$. The affine map from the reference square is $F_{l(\hat{x}, \hat{y})} = (x_i + h_x \hat{x}, y_j + h_y \hat{y})$.

The physical basis functions are $\varphi_r^l = \varphi_r \circ F_l^{-1}$ for $r = 1, \dots, 4$. The physical local stiffness matrix is $K_{rs}^l = \int_{K_l} \nabla \varphi_s^l \cdot \nabla \varphi_r^l \, dx \, dy$.

By change of variables,

$$K^l = \begin{pmatrix} h_y \\ h_x \end{pmatrix} K_x^{\text{ref}} + \begin{pmatrix} h_x \\ h_y \end{pmatrix} K_y^{\text{ref}}.$$

If $h_x = h_y$, then $K^l = K^{\text{ref}}$. This scale cancellation is specific to the pure Laplace stiffness matrix in dimension 2.

6 Local Load Vector

The local load vector is $F_r^l = \int_{K_l} f \varphi_r^l \, dx \, dy$. On the reference element,

$$F_r^l = h_x h_y \int_{[0,1]^2} f(F_{l(\hat{x}, \hat{y})}) \varphi_{r(\hat{x}, \hat{y})} d\hat{x} d\hat{y}.$$

If f is simple, compute this exactly. Otherwise use quadrature.

7 Assembly Method

For every cell K_l , define the local-to-global map $I_l : \{1, 2, 3, 4\} \rightarrow \{1, \dots, N_v\}$, where $I_{l(r)}$ is the global index of local node r .

Assembly means adding local contributions into global entries:

$$A_{I_{l(r)}, I_{l(s)}} += K_{rs}^l, \quad F_{I_{l(r)}} += F_r^l.$$

Precise algorithm:

Initialize $A = 0$ and $F = 0$.

```
For each element K_l:
  get global indices I_l(1), ..., I_l(4)
  compute K^l and F^l
  for r = 1, ..., 4:
    F[I_l(r)] += F^l[r]
  for r = 1, ..., 4 and s = 1, ..., 4:
    A[I_l(r), I_l(s)] += K^l[r, s]
```

This works because global basis functions are obtained by gluing local basis functions. A global entry A_{ij} is the sum of the local integrals over all cells touching both nodes i and j .

8 Dirichlet Boundary Conditions

If $u = 0$ on $\partial\Omega$, boundary vertex values are fixed to zero. The clean variational method is to assemble only the interior unknowns: $V_{h,0} = \text{span} \{\Phi_i : a_i \text{ is interior}\}$. Equivalently, one may assemble the full system and replace each boundary row by an identity row with $F_i = 0$.

If $u = g$ on $\partial\Omega$, write $u_h = w_h + g_h$, where $w_h \in V_{h,0}$ and g_h is the Q_1 interpolant of the boundary data. Then solve

$$\int_{\Omega} \nabla w_h \cdot \nabla v_h dx = \int_{\Omega} f v_h dx - \int_{\Omega} \nabla g_h \cdot \nabla v_h dx \quad \forall v_h \in V_{h,0}.$$

In matrix form, after splitting interior and boundary indices, $A_{II}U_I = F_I - A_{IB}G_B$.

9 Mass Matrix Reminder

For $-\Delta u + u = f$, the bilinear form is $a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v dx + \int_{\Omega} uv dx$. Add the local mass matrix M^l to the local stiffness matrix.

On the reference square, $M_{ij}^{\text{ref}} = \int_K \varphi_j \varphi_i dx dy$, and

$$M^{\text{ref}} = \begin{pmatrix} \frac{1}{9} & \frac{1}{18} & \frac{1}{18} & \frac{1}{36} \\ \frac{1}{18} & \frac{1}{9} & \frac{1}{36} & \frac{1}{18} \\ \frac{1}{18} & \frac{1}{36} & \frac{1}{9} & \frac{1}{18} \\ \frac{1}{36} & \frac{1}{18} & \frac{1}{18} & \frac{1}{9} \end{pmatrix}.$$

On K_l , $M^l = h_x h_y M^{\text{ref}}$. For $-\Delta u + u = f$, assemble $A^l = K^l + M^l$.

10 Minimal Checklist

1. Local ordering: $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$.
2. Local basis: $\varphi_1 = (1-x)(1-y)$, $\varphi_2 = x(1-y)$, $\varphi_3 = (1-x)y$, $\varphi_4 = xy$.
3. Rectangle formula: $K^l = \begin{pmatrix} h_y \\ h_x \end{pmatrix} K_x^{\text{ref}} + \begin{pmatrix} h_x \\ h_y \end{pmatrix} K_y^{\text{ref}}$.
4. Assembly: add K_{rs}^l into $A_{I_{l(r)}, I_{l(s)}}$ and F_r^l into $F_{I_{l(r)}}$.
5. Dirichlet: keep only interior nodes, or impose boundary rows after full assembly.